

can you describe entire project context in very depth all modules in depth and step by step as flow oriented

with out any schema name

Go----Below is the **complete CMCMIS project context** in a **deep, flow-oriented way**, without mentioning any database schema/table names.

CMCMIS Project Context

CMCMIS means **Computerized Maintenance & Calibration Management Information System**.

It is a production-grade web application for managing the complete lifecycle of instruments, equipment, calibration, repair, registration, scheduling, inquiry, reporting, procurement, and administrative workflows inside a technical organization such as **ISRO SAC**.

The main goal is to replace manual, paper-based, scattered, or legacy workflows with a centralized web-based system where every user can perform their role with minimum clicks, proper permissions, real-time tracking, and reliable data.

The system is not just a dashboard. It is an operational workflow platform.

It manages:

Equipment → Job Request → Approval/Assignment → Job Card → Calibration/Repair Work → Completion → History → Reports

For new equipment:

New Equipment Registration Request → Procurement Verification → Physical Verification → Equipment ID Assignment → Equipment Master Entry → Calibration Schedule

1. User Roles and Their Responsibilities

1. Normal User

This is the general employee or scientist who uses equipment and submits requests.

Normal user can:

- Login using employee ID for now.
- View personal dashboard.
- Submit new calibration request.
- Submit new repair request.
- Submit new equipment registration request.
- Track their own job request status.
- View their own equipment or division-related equipment.
- Use inquiry to search allowed information.

- See status updates but not modify technical job card details.

Typical flow:

User logs in

- Dashboard opens
- User clicks New Service Request
- Selects Calibration / Repair / Registration
- Fills equipment and submitted-by details
- Accepts terms
- Submits request
- Tracks request status later

2. Lab Engineer

This is the technical person who actually performs calibration, repair, inspection, or maintenance work.

Lab engineer can:

- View assigned jobs.
- Open job cards.
- Enter observations.
- Update calibration details.
- Update repair details.
- Add test results.
- Add faults, actions taken, and remarks.

- Mark work progress.
- Attach reports if required.
- Complete job card after work is done.

Typical flow:

Lab engineer logs in

- Sees assigned job cards
- Opens one job card
- Reviews request and equipment details
- Performs calibration/repair
- Fills technical observations
- Updates status
- Completes job card

3. Lab In-charge

This user supervises work within a lab or division.

Lab in-charge can:

- View requests coming to their lab/division.
- Approve or review job requests.
- Assign engineers.
- Monitor workload.
- Track pending, overdue, and completed jobs.
- Review calibration/repair progress.
- Ensure job closure.

- View reports for their division.

Typical flow:

Lab in-charge logs in

- Reviews pending requests
- Approves request
- Assigns engineer
- Monitors job card progress
- Reviews completed work
- Closes or forwards if needed

4. Admin

Admin controls system-level configuration.

Admin can:

- Manage users.
- Manage roles.
- Manage access rights.
- Manage master data.
- View all equipment and job records.
- Manage divisions, products, vendors, tasks, checklist data, and other system configuration.
- Generate reports.
- Audit the system.

Typical flow:

Admin logs in

- Opens admin panel
- Updates master data / permissions / roles
- Monitors full system
- Generates reports

5. View-only User

This user can only inspect records.

View-only user can:

- Login.
- View allowed modules.
- Search and inspect records.
- Cannot create, edit, delete, approve, assign, or submit.

Typical flow:

View-only user logs in

- Opens dashboard/inquiry
- Searches information
- Views status/history only

2. Authentication and Access Flow

The current project uses a **temporary employee-ID login system** for development.

Later, it is designed to support **SSO**, meaning users will log in through the organization's official identity system.

Current flow:

User enters Employee ID

- Backend verifies employee exists
- Backend loads user role
- Backend loads permissions
- Secure session is created
- Frontend receives user profile and allowed modules
- Sidebar is shown according to permissions

Future SSO flow:

User clicks Organization SSO Login

- Organization verifies identity
- CMCMIS receives verified employee identity
- CMCMIS maps identity to internal role
- CMCMIS loads permissions
- Secure session is created
- User enters system

Important concept:

SSO decides: Who is this user?

CMCMIS decides: What can this user do?

So even after SSO, CMCMIS still needs its own role and permission logic.

3. Main Application Layout Flow

After login, every user sees a common structure:

Sidebar navigation

Topbar search

Notification area

User profile

Main content area

Sidebar modules are shown based on permission.

Example:

Dashboard

Job Requests

Equipment

Inquiry

Reports

Admin

If user does not have permission for a module, that module should not appear in sidebar.

But security must always be enforced from backend also.

Frontend hiding is only for better user experience.

4. Dashboard Module

The dashboard is the first landing page after login.

Its purpose is to give the user a quick operational overview.

Normal User Dashboard

The normal user dashboard shows:

- Active requests
- Requests in progress
- Completed requests this month
- Equipment due for calibration
- My service requests
- My equipment
- Quick links
- Important notes
- Help/contact information

Flow:

User logs in

→ Dashboard loads

→ Backend identifies logged-in user

→ Backend fetches only user-scoped data

→ Dashboard cards are filled with real values

→ User can quickly submit new request or track existing request

Dashboard card logic:

Active Requests

→ Requests submitted by user that are not completed

In Progress

→ Requests where work has started but not completed

Completed This Month

→ Requests completed in current month

Due for Calibration

→ User/division-related equipment due soon

Dashboard should never show fake numbers. If no data exists, it should show zero or empty state.

5. Job Requests Module

This module manages all service requests.

A job request is the starting point for:

- Calibration
- Repair
- Registration

Job Request Listing Page

This page shows all job requests visible to the current user.

UI contains:

- Search
- Type filter
- Status filter

- Advanced filters
- Export button
- New Job Request button
- Paginated table
- Mobile-friendly card view

Flow:

User opens Job Requests

- Backend checks permission
- Backend checks user role
- Backend applies row-level visibility
- Backend returns only allowed requests
- Frontend displays request list

Role-wise visibility:

Normal User

- Sees own submitted requests

Lab Engineer

- Sees assigned/related requests

Lab In-charge

- Sees division/lab requests

Admin

→ Sees all requests

View-only User

→ Sees scoped read-only records

Job Request Status Meaning

The system derives status based on workflow condition.

Possible statuses:

Submitted

Pending Approval

Approved

In Progress

Completed

Rejected / Closed if supported later

Example status flow:

User submits request

→ Pending Approval

→ Lab in-charge approves

→ Approved

→ Engineer starts work

→ In Progress

→ Engineer completes work

- Completed
- Lab closes record
- Closed

6. New Job Request Form

This is one of the most important flows.

Route:

/job-requests/new

Opened from:

Dashboard → New Service Request

Job Requests → New Job Request

The form supports:

Calibration

Repair

Registration

General Form Flow

User opens form

- Selects job category
- Selects job type
- Provides equipment details
- Adds accessories if needed
- Submitted-by details are autofilled
- User enters complaint/requirement

- User accepts terms and conditions
- Submit button becomes enabled
- Backend saves request
- Success popup appears

Form Sections for Calibration / Repair

Section 1: Job Type Selection

User selects:

Job Category

- T&ME or F&PE

Job Type

- Calibration or Repair

Section 2: Equipment Details

For calibration or repair, equipment should already exist.

Flow:

User enters equipment ID/name/model/serial

- Backend searches equipment records
- User selects equipment
- Form autofills equipment details

Autofilled fields:

Equipment name

Make/manufacturer display

Model number

Serial number

Equipment type

Description/options

Division

Calibration due date if needed

Section 3: Accessories

User can add optional accessories.

Flow:

User selects accessory type

→ Enters accessory name

→ Enters serial number

→ Clicks Add Accessory

→ Accessory appears in added list

→ User can delete it before submit

Accessory examples:

Current Probe

Detector

Noise Source

Plugin Sensor

Power Supply

Adapter

Cable

Antenna

Others

Section 4: Submitted By

This section is autofilled using the logged-in employee.

Fields:

Name

Employee ID

Designation

Email

Lab phone

Room phone

Division

Subsystem

Project

Flow:

Backend reads logged-in employee

→ Fetches employee details

→ Fetches division details

→ Frontend autofills form

If user manually enters another employee ID, backend validates whether that employee exists and is active.

Section 5: Terms and Conditions

Submit button remains disabled until all terms are accepted.

Flow:

User checks all six terms

→ Accepted count reaches 6/6

→ Submit button becomes active

Terms acceptance is currently validation-only because no confirmed audit storage flow exists yet.

Submit Flow

When user submits:

Frontend validates fields

→ Backend validates again

→ Backend generates request number

→ Backend starts transaction

→ Saves main request

→ Saves accessories if any

→ Saves project mapping if any

→ Commits transaction

→ Sends success response

→ Frontend shows success popup

If any step fails:

Rollback transaction

→ No partial data saved

→ User receives safe error message

7. Registration Mode / Add Equipment Flow

This is the special flow for new equipment registration.

Opened from:

Equipment page → Add Equipment

Route:

/job-requests/new?type=registration

Important decision:

Add Equipment does not directly create equipment master record right now.

It creates an equipment registration request.

Reason:

New equipment should be verified first.

Equipment ID should be assigned after successful verification.

Registration Flow

User clicks Add Equipment

→ Registration form opens

→ Job Type is pre-selected as Registration

→ User fills new equipment details

→ User fills procurement details

→ User fills submitted-by details

- User accepts registration-specific terms
- User submits request
- Backend saves registration request
- Equipment ID is assigned later after verification

Registration Form Sections

Section 1: Job Type Selection

Job Category

- T&ME or F&PE

Job Type

- Registration

Helper text:

Pre-selected for equipment registration

Section 2: Equipment Details

Since equipment does not exist yet, equipment ID is not required.

Fields:

Equipment Name

Make

Model No.

Serial No.

Equipment Type

Options / Description

Message shown:

New Equipment Registration: Equipment ID will be auto-generated after successful registration and verification.

Section 3: Accessories

Same as normal request.

User can add manuals, probes, cables, detectors, sensors, adapters, etc.

Section 4: Procurement Details

Visible only in registration mode.

Fields:

PO Number

PO Date

MIVR/SRV Number

MIVR/SRV Date

Line Item Code

Cost

Cost Currency

Warranty Period

Purpose:

To verify that equipment was procured properly before registration.

Section 5: Submitted By

Autofilled employee details.

Section 6: Registration Terms and Conditions

Registration-specific terms include:

Procurement and equipment details are accurate.

Registration begins after PO and MIVR/SRV verification.

Equipment ID will be assigned after verification.

Accessories/manuals/certificates will be provided.

Warranty documentation is maintained.

Equipment will enter calibration schedule after registration.

Submit button activates only after all six terms are accepted.

8. Equipment Module

Route:

/equipment

Purpose:

Manage and track all equipment inventory.

The equipment page shows real inventory data.

UI contains:

Search by ID, name, make, model, serial

Type filter

Status filter

Advanced filters

Export button

Add Equipment button

Paginated table

Mobile card layout

Equipment Listing Flow

User opens Equipment page

→ Backend checks permission

→ Backend checks user role

→ Backend applies equipment scope

→ Backend fetches equipment records

→ Frontend displays rows

Role-wise visibility:

Normal User

→ Sees equipment related to own division or own requests

Lab Engineer

→ Sees assigned/allowed equipment

Lab In-charge

→ Sees equipment in assigned divisions

Admin

→ Sees all equipment

View-only

→ Sees scoped read-only equipment

Equipment Status

Equipment status is not treated as a fake static field. It is derived from available operational information.

Possible statuses:

Operational

Under Calibration

Under Repair

Calibration Due

Calibration Overdue

PM Due

PM Overdue

End Of Support

Example derivation:

If active calibration job exists

→ Under Calibration

If active repair job exists

→ Under Repair

If calibration due date has passed

→ Calibration Overdue

If calibration due date is within next 30 days

→ Calibration Due

Otherwise

→ Operational

Add Equipment Flow

User clicks Add Equipment

→ Redirect to registration mode

→ User submits registration request

→ Equipment master creation happens later after verification

9. Inquiry Module

Route:

/inquiry

Purpose:

Search and lookup information across the system.

Inquiry is a read/search module.

Tabs:

Vendor

Product

Job Card Status

Instrument Lookup

Inquiry Main Flow

User opens Inquiry

→ Selects tab

→ Enters search text

→ Backend checks permission

→ Backend fetches scoped data

→ Frontend displays results

Vendor Tab

Purpose:

Search vendor/supplier/manufacture/service-provider information.

Expected fields:

Vendor ID

Name

Type

Contact

Email

Address

Important:

Vendor fields must only be shown if verified source columns are available.

No fake vendor data should be displayed.

If vendor source fields are not confirmed:

Show professional message:

Vendor inquiry requires verified vendor master configuration.

Product Tab

Purpose:

Search product or instrument type information.

Fields:

Product ID

Product Name

Description

Category

State

Category may be derived from product flags.

Example categories:

Instrument

T&ME

Product

General

Job Card Status Tab

Purpose:

Track job card progress.

Fields:

Job ID

Equipment

Status

Assigned Engineer

Submitted Date

Progress

Flow:

User searches job/equipment

→ Backend fetches job card details

→ Backend derives status and progress

→ Frontend displays progress bar

Progress logic:

Completed → 100%

Pending → 0%

In Progress → calculated using dates or safe progress estimate

Instrument Lookup Tab

Purpose:

Search equipment/instrument details quickly.

Fields:

Equipment ID

Name

Division

Location

Status

Last Calibration

Last calibration should come from actual historical calibration completion information, not fake dates.

10. Job Card Module

This module is the next major operational module after job requests.

A job card represents actual lab work after a request is approved.

Job Card Lifecycle

Request Submitted

→ Request Approved

→ Job Card Created

→ Engineer Assigned

→ Work Started

→ Observations Entered

→ Calibration/Repair Details Filled

→ Work Completed

→ Report Uploaded

→ Job Closed

Lab Engineer Flow

Engineer logs in

→ Opens assigned job card

→ Reviews equipment/request details

→ Starts work

→ Enters calibration or repair data

→ Adds observations/checklist

→ Updates status

→ Completes job card

Lab In-charge Flow

Lab in-charge reviews open job cards

→ Checks engineer workload

→ Assigns/reassigns engineer

→ Reviews completed work

→ Closes job card

Job Card Types

Job card can represent:

Calibration

Repair

Inspection

Preventive maintenance

Registration verification

Information captured

Job start date

Job end date

Engineer assigned

Equipment received date

Fault details

Repair action taken

Calibration result

Observations

Spare/device used

Remarks

Report upload

Status history

11. Scheduling Module

This module manages planned calibration and preventive maintenance.

Purpose:

Ensure equipment is calibrated or maintained before due date.

Scheduling Flow

System reads equipment due dates

- Generates schedule plan
- Lab reviews upcoming tasks
- Engineer performs scheduled calibration/PM
- Job card is created if needed
- Completion updates future due date

Views:

Calendar view

Table view

Due soon

Overdue

Plan vs actual

Important dashboard indicators:

Calibration due in next 30 days

Preventive maintenance due

Overdue equipment

Upcoming scheduled tasks

12. Procurement Module

Procurement supports equipment purchase, spare parts, vendor relation, warranty, and purchase tracking.

Purpose:

Track procurement-related information linked to equipment and repair/calibration workflows.

Procurement Flow

Equipment purchased

- PO/MIVR/SRV details recorded
- Equipment registration request submitted
- Verification done
- Equipment ID assigned
- Equipment enters inventory

For repair:

Fault detected

- Spare required
- Spare availability checked
- Purchase/indent details added
- Repair completed

Data handled:

PO number

PO date

Vendor

Cost

Budget/line item

Warranty

Spare part details

Inventory balance

AMC/contract details

13. Reports Module

Reports are used by admin, lab in-charge, and management.

Purpose:

Convert operational data into useful summaries.

Possible reports:

Calibration due report

Overdue equipment report

Pending job requests

Completed job cards

Engineer workload

Lab utilization

AMC report

Purchase order report

Equipment history report

Repair frequency report

Division-wise workload

Report Flow

User selects report type

→ Applies filters

- Backend fetches real data
- Frontend shows table/graphs
- User exports PDF/Excel if permitted

Filters:

Date range

Division

Equipment type

Engineer

Status

Job type

Calibration due range

Export must require permission.

14. Admin Module

Admin module manages master data and access control.

Purpose:

Control system configuration.

Admin can manage:

Employees

Roles

Permissions

Divisions

Equipment types

Products

Vendors

Tasks

Checklists

Spare/device types

Modules/menu visibility

Access rights

Admin Flow

Admin logs in

- Opens Admin panel
- Selects master category
- Adds/updates/deactivates records
- Changes permissions
- System updates menus/API access

Important:

Admin actions must be audited later.

Admin must not accidentally corrupt legacy data.

Soft-disable is safer than hard-delete.

15. Permission Flow Across System

Every module follows the same permission flow.

User calls API

- Backend verifies session

- Backend loads role
- Backend checks permission
- Backend applies row-level scope
- Backend returns only allowed data

Permission actions:

View

Create

Update

Delete

Print/Export

Example:

User may view equipment

But may not add equipment

So Equipment page is visible

But Add Equipment button is hidden

Backend must still block unauthorized creation even if frontend button is hidden.

16. Global Search Flow

Topbar search is meant to search across major entities.

Possible search areas:

Equipment

Job requests

Job cards

Vendors

Products

Employees

Flow:

User types in topbar

→ Backend searches allowed modules

→ Results grouped by type

→ User clicks result

→ Redirects to relevant detail/list page

This should be implemented carefully later because global search must respect permissions.

17. Notification Flow

Notification system can be added later.

Possible notifications:

Request approved

Request rejected

Engineer assigned

Job completed

Calibration due soon

Equipment overdue

Request needs correction

Registration verified

Report uploaded

Flow:

System event occurs

→ Notification generated

→ User sees bell indicator

→ User opens notification

→ Redirects to relevant page

18. Data Integrity Flow

Because this is a live-user production system, data consistency is critical.

For parent-child operations, use transaction.

Example new job request:

Generate request number

→ Insert main request

→ Insert accessories

→ Insert project mapping

→ Commit

If accessory insert fails:

Rollback everything

→ No half-created request

This prevents corrupted workflows.

19. No Mock Data Rule

The system must not show fake operational data.

Allowed:

UI labels

Status color mappings

Static help text

Temporary controlled dropdown constants if DB source is unavailable

Not allowed:

Fake equipment rows

Fake job request rows

Fake dashboard counts

Fake vendors

Fake job cards

Fake calibration dates

If database has no records:

Show empty state

Show zero count

Show “No records found”

20. Current Development Priority

The ideal implementation order is:

1. Backend foundation

2. Database connection
3. Auth/RBAC
4. Dashboard
5. Job Requests listing
6. Equipment listing
7. Inquiry
8. New Job Request form
9. Registration mode
10. Job Card lifecycle
11. Scheduling
12. Reports
13. Admin/master data
14. Real SSO integration
15. Deployment and monitoring

21. End-to-End Example Flow

Calibration Request

User logs in

- Opens dashboard
- Clicks New Service Request
- Selects category
- Selects Calibration
- Searches existing equipment

- Equipment details autofill
- Adds accessories if needed
- Submitted-by details autofill
- Enters complaint/requirement
- Accepts T&C
- Submits request
- Request appears in Job Requests list
- Lab in-charge approves
- Job card created
- Engineer assigned
- Engineer performs calibration
- Calibration result entered
- Job completed
- User tracks completed status
- Report available

Repair Request

User logs in

- Creates Repair request
- Selects existing equipment
- Enters fault/complaint
- Adds accessories if needed
- Submits

- Lab reviews
- Engineer diagnoses fault
- Spare/repair action added
- Repair completed
- Equipment returned
- Request closed

Equipment Registration

User clicks Add Equipment

- Registration form opens
- Enters new equipment details
- Enters procurement details
- Adds accessories/manuals if needed
- Submitted-by details autofill
- Accepts registration T&C
- Submits registration request
- Lab verifies PO/MIVR/SRV and physical equipment
- Equipment ID assigned
- Equipment added to inventory
- Calibration schedule created

22. Final Project Vision

CMCMIS should become a complete technical operations platform where:

Users submit requests easily.

Lab engineers process work accurately.

Lab in-charges monitor and assign efficiently.

Admins control roles and master data.

Management gets reliable reports.

Equipment history remains traceable.

Calibration and repair workflows become transparent.

The final system should be:

Fast

Secure

Permission-aware

Minimal-click

Responsive

Database-driven

Production-ready

Government-grade

Maintainable

Auditable

In one line:

CMCMIS is a role-based, real-data-driven operational system that manages the complete lifecycle of equipment, calibration, repair, registration, job cards, inquiry,

scheduling, procurement, and reporting for a high-precision technical organization.

Saved. From now onwards, I'll follow this response style for your CMCMIS/project explanations:

Structured step-by-step format

Memory-based project context

Table-wise explanations

Flowcharts

Diagrams / architecture visuals

Concept relationships

Simple explanation + deep technical details

Real-world examples

Module connections

Summarized notes at the end

Beginner-to-advanced friendly

Easy to revise

Below is your **entire CMCMIS chat context** in one structured block.

***** CMCMIS COMPLETE PROJECT CONTEXT BLOCK *****

0. Core Project Identity

Project Name:

CMCMIS — Computerized Maintenance & Calibration Management Information System

Project Type:

Production-grade web-based calibration, maintenance, repair, equipment registration, job card, inquiry, procurement, scheduling, reporting, and admin management system.

Target Environment:

High-precision technical/government engineering environment like ISRO SAC.

Core Goal:

Build a real-data-driven, role-based, secure, minimal-click, professional,

industry-grade operational platform for engineers, lab staff, admins, and users.

The system is not just a dashboard. It is a complete operational workflow system.

Equipment

- Job Request
- Approval / Assignment
- Job Card
- Calibration / Repair / Registration Work
- Completion
- History
- Reports

For new equipment:

New Equipment Registration Request

- Procurement Verification
- Physical Verification
- Equipment ID Assignment
- Equipment Master Entry
- Calibration Schedule

Reference project context was also captured in the uploaded markdown/project context file.

1. Fixed Technical Stack

Layer	Technology	Purpose
Frontend	React.js	Build user interface
Frontend Language	JavaScript	No TypeScript for now
Build Tool	Vite	Fast frontend dev server
Styling	Tailwind CSS	Responsive professional UI
Backend	Node.js	Runtime
Backend Framework	Express.js	REST API server
Database	MySQL	Existing relational database
Auth Session	JWT + HttpOnly Cookie	Secure login/session

Layer	Technology	Purpose
Database Driver	mysql2/promise	Fast SQL queries and pooling

Important Rule

Do not create a new database schema.

Do not invent new tables.

Do not invent new columns.

Use the existing MySQL database only.

All backend work must match the existing table and column structure.

2. Main Development Philosophy

Simple Explanation

The system should feel easy for users, but internally it must be production-grade.

Deep Technical Meaning

Every module should follow:

Frontend Page

→ API File

→ Route

→ Middleware

→ Controller

→ Service

→ Repository

→ MySQL Query

→ Mapper

→ Response

→ UI Render

Architecture Flow

React UI

↓

Axios API Layer

↓

Express Route

↓

Auth Middleware

↓

RBAC Middleware

↓

Controller

↓

Service Logic

↓

Repository SQL

↓

Existing MySQL DB

3. User Roles

Role	Main Responsibility	Access Style
Normal User	Submit and track requests	Own requests and scoped equipment
Lab Engineer	Work on assigned job cards	Assigned technical work
Lab In-charge	Approve, assign, monitor	Division/lab-level control
Admin	Manage system and master data	Full system access
View-only User	Inspect records only	Read-only scoped access

Role Flow

User logs in

→ Role is loaded

→ Permissions are loaded

→ Sidebar modules are filtered

→ API access is checked

→ Data scope is applied

Important Security Rule

Frontend hiding is only UX.

Backend permission checking is mandatory.

Example:

If user cannot export Equipment:

- Export button should be hidden in UI
 - Backend export API must still block access
-

4. Authentication and SSO-Ready Flow

Current Auth Decision

You decided to use:

Temporary Employee-ID Login

+

JWT/Session

+

RBAC

+

Future SSO Adapter

Current Login Flow

User enters Employee ID

- Backend checks employee
- Backend loads role
- Backend loads permissions
- JWT session is created
- Cookie is set as HttpOnly
- Frontend loads dashboard/sidebar

Future SSO Flow

User clicks Organization SSO

- Organization verifies identity
- CMCMIS receives verified employee identity
- CMCMIS maps identity to role
- CMCMIS loads permissions
- Session is created
- User enters app

Important Concept

SSO answers:

Who is this user?

CMCMIS RBAC answers:

What can this user do?

Development Login IDs

Employee ID Role

user7990	Normal User
labin7990	Lab In-charge
labeng7990	Lab Engineer
admin001	Admin SUPER
viewer7990	View-only User

5. Current Blocking Issues You Found

Issue Summary

Issue	Problem	Impact	Fix Area
Permissions empty	Role permissions missing	Sidebar empty	DB permission/fallback config
F5 needed after login	Dashboard hook not triggered	Bad UX	Frontend auth/dashboard hook
Sidebar empty	Permission array empty	No navigation	Sidebar + permissions
Dashboard controller broken	Methods commented/missing	APIs fail	Dashboard controller/routes
Empty DB data	No operational records	Zero dashboard	Real seed/test data or empty states

Fix Strategy

1. Fix permission loading first.
2. Ensure fallback permissions in development.
3. Fix sidebar empty state.
4. Fix dashboard fetch after login.

5. Restore dashboard controller methods.
6. Fix job request form endpoints.
7. Add safe empty states if DB has no records.

6. Global App Layout

Current/Desired Layout

Left Sidebar

Top Header

Global Search

Notification Bell

User Profile

Main Content Area

Sidebar Modules

Dashboard

Job Requests

Job Cards

Equipment

Schedule

Procurement

Inquiry

Reports

Admin

UI Rule

Role-based modules should appear only when user has permission.

Responsive Behavior

Device UI Behavior

Desktop Full sidebar + full tables

Tablet Collapsible sidebar + 2-column grids

Mobile Drawer sidebar + card layouts

7. Dashboard Module

Purpose

The dashboard gives a quick operational overview after login.

Dashboard Shows

Active Requests

In Progress Requests

Completed This Month

Calibration Due

My Service Requests

My Equipment

Quick Links

Important Notes

Help Section

Flow

User logs in

→ Dashboard opens

→ Backend identifies user

→ Backend applies user scope

→ Real DB data is fetched

→ Cards/tables render

Dashboard Logic

Card	Meaning
Active Requests	Submitted but not completed
In Progress	Work started but not completed
Completed This Month	Finished in current month
Due for Calibration	Equipment due soon

Important Rule

No fake dashboard numbers.

If database has no records, show 0 or empty state.

8. Job Requests Listing Module

Route

/job-requests

Purpose

Manage and track calibration, repair, and registration requests.

UI Elements

Search

Type Filter

Status Filter

Advanced Filters

Export

New Job Request Button

Paginated Table

Mobile Cards

Flow

User opens Job Requests

→ Backend checks permission

→ Backend checks role

→ Backend applies row-level scope

→ Backend returns allowed requests

→ Frontend displays table/cards

Role-wise Visibility

Role	Data Visibility
Normal User	Own submitted requests
Lab Engineer	Assigned/related requests
Lab In-charge	Division/lab requests
Admin	All requests
View-only	Scoped read-only

Status Flow

Submitted

→ Pending Approval

→ Approved

→ Assigned

→ In Progress

→ Completed

→ Closed

Important Derived Fields

Field	Source Logic
--------------	---------------------

Status	Derived from request/job progress
--------	-----------------------------------

Priority	Derived from request age
----------	--------------------------

Display ID	Generated for UI from real request number/date
------------	--

9. New Job Request Form

Route

/job-requests/new

Opened From

Dashboard → New Service Request

Job Requests → New Job Request

Supported Types

Calibration

Repair

Registration

General Flow

User opens form

→ Selects job category

→ Selects job type

→ Enters/selects equipment

→ Adds accessories if needed

→ Submitted-by autofills

→ User enters complaint/remarks

- User accepts T&C
- Submit button activates
- Backend saves transactionally
- Success popup appears

Sections for Calibration/Repair

Section	Purpose
1. Job Type Selection	Select T&ME/F&PE and Calibration/Repair
2. Equipment Details	Search existing equipment and autofill
3. Accessories	Add optional accessories
4. Submitted By	Autofill employee details
5. Complaint / T&C	Complaint, remarks, terms acceptance

Equipment Search Flow

User types equipment ID/name/model/serial

- Backend searches equipment
- User selects result
- Form autofills equipment fields

Submit Flow

Frontend validation

- Backend validation
- Generate request number
- Begin transaction
- Insert main request
- Insert accessories
- Insert project mapping
- Commit
- Return success

If error:

Rollback transaction

- No partial save
- Return safe error

10. Equipment Registration / Add Equipment Flow

Route

/job-requests/new?type=registration

Opened From

Equipment Page → Add Equipment

Important Decision

Add Equipment does NOT directly create final equipment record.

It creates an equipment registration request.

Why

New equipment should be verified first.

Equipment ID should be assigned after verification.

Registration Flow

User clicks Add Equipment

→ Registration mode opens

→ Job Type is pre-selected as Registration

→ User enters new equipment details

→ User enters procurement details

→ Submitted-by autofills

→ User accepts registration T&C

→ Request is submitted

→ Equipment ID assigned later after verification

Registration Sections

Section	Fields
Job Type	Category + Registration
Equipment Details	Name, Make, Model, Serial, Type, Description
Accessories	Optional accessories/manuals/probes
Procurement Details	PO, MIVR/SRV, Line Item, Cost, Warranty
Submitted By	Employee details
Terms	Six registration-specific checkboxes

Registration Success Message

Registration request submitted successfully.

Equipment ID will be assigned after verification.

11. Equipment Module

Route

/equipment

Purpose

Manage and track all equipment inventory.

UI Elements

Search

Type Filter

Status Filter

Advanced Filters

Export

Add Equipment

Paginated Table

Mobile Cards

Equipment Flow

User opens Equipment

→ Backend checks permission

→ Backend checks role

→ Backend applies equipment scope

→ Real equipment data is returned

→ UI displays rows/cards

Status Types

Operational

Under Calibration

Under Repair

Calibration Due

Calibration Overdue

PM Due

PM Overdue

End Of Support

Status Derivation Example

Active calibration job exists

→ Under Calibration

Calibration due date passed

→ Calibration Overdue

Calibration due within 30 days

→ Calibration Due

No issue

→ Operational

Add Equipment Flow

Click Add Equipment

→ Redirect to Registration Mode

→ Submit registration request

→ Equipment created later after verification

12. Inquiry Module

Route

/inquiry

Purpose

Search and lookup system information.

Tabs

Vendor

Product

Job Card Status

Instrument Lookup

Inquiry Flow

User opens Inquiry

- Selects tab
- Searches keyword
- Backend checks permission
- Backend applies user scope
- Results are returned

Tab Details

Tab	Purpose
Vendor	Search vendor/supplier/manufacture
Product	Search product/instrument types
Job Card Status	Track job progress
Instrument Lookup	Search equipment/instrument details

Important Rule

If vendor columns/source are not verified,
do not show fake vendor data.
Show configured-false/professional message.

13. Job Cards Module

Purpose

Job Cards represent actual lab work after request approval.

Lifecycle

- Request Submitted
- Request Approved
 - Job Card Created
 - Engineer Assigned
 - Work Started
 - Observations Entered
 - Calibration/Repair Details Filled
 - Work Completed

→ Report Uploaded

→ Job Closed

Lab Engineer Flow

Engineer logs in

→ Opens assigned job card

→ Reviews request/equipment

→ Starts work

→ Enters observations

→ Adds calibration/repair details

→ Uploads report

→ Completes job

Lab In-charge Flow

In-charge reviews jobs

→ Assigns engineer

→ Monitors progress

→ Reviews completed work

→ Closes/approves final status

Captured Data

Start date

End date

Assigned engineer

Equipment received date

Fault details

Repair action

Calibration result

Observations

Spares/device used

Remarks

Reports

Status history

14. Schedule Module

Purpose

Manage preventive maintenance and calibration planning.

Flow

System reads equipment due dates

→ Schedule plan is generated

→ Lab reviews due list

→ Engineer performs calibration/PM

→ Job card created if needed

→ Completion updates future due date

Views

Calendar View

Table View

Due Soon

Overdue

Plan vs Actual

Schedule Types

Calibration Schedule

Preventive Maintenance Schedule

15. Procurement Module

Purpose

Track procurement, PO, vendor, warranty, spare parts, and linked registration.

Procurement Flow

Equipment purchased

→ PO/MIVR/SRV recorded

→ Registration request created

→ Verification done

→ Equipment ID assigned

→ Equipment enters inventory

Repair Procurement Flow

Fault detected

→ Spare required

→ Spare availability checked

→ Purchase/indent added

→ Repair completed

Data Handled

PO number

PO date

Vendor

Cost

Line item/budget

Warranty

Spare parts

Inventory balance

AMC/contract details

16. Reports Module

Purpose

Convert operational data into management-ready insights.

Report Types

Calibration Due Report

Overdue Equipment Report

Pending Job Requests

Completed Job Cards

Engineer Workload

Lab Utilization

AMC Report

Purchase Order Report

Equipment History Report

Repair Frequency Report

Division-wise Workload

Report Flow

User selects report

→ Applies filters

→ Backend fetches real data

→ Charts/tables render

→ Export PDF/Excel if permitted

Filters

Date range

Division

Equipment type

Engineer

Status

Job type

Due range

17. Admin Module

Purpose

Control system configuration and master data.

Admin Can Manage

Employees

Roles

Permissions

Divisions

Equipment Types

Products

Vendors

Tasks

Checklists

Spare/device types

Modules/Menu

Access Rights

Admin Flow

Admin logs in

- Opens Admin
- Selects master category
- Adds/updates/deactivates records
- Updates permissions
- Sidebar/API access changes

Admin Rule

Soft-disable is safer than hard-delete.

Critical admin actions should be confirmed and audited.

18. Permissions and RBAC Flow

Permission Actions

View

Create

Update

Delete

Print / Export

Global Permission Flow

User calls API

- Backend verifies session
- Backend loads role
- Backend checks permission
- Backend applies row-level scope
- Backend returns allowed data only

Example

User has Equipment View permission

- Equipment page visible

User does not have Equipment Create permission

- Add Equipment button hidden

→ Backend create API also blocked

19. Database Connection Status

You tested backend and saw:

[DB]  Connected to MySQL database: cmcmis_redev @ localhost:3306

[SERVER]  CMCMIS API running on http://localhost:5000

Meaning

Backend is running.

Database connection is successful.

Frontend origin is configured.

Auth mode is DEV_EMPLOYEE.

SSO is disabled for now.

Next Test Flow

Test /api/health

→ Test /api/auth/dev-login

→ Test /api/auth/me

→ Start frontend

→ Login with user7990

→ Verify sidebar

→ Verify dashboard

20. No Mock Data Rule

Allowed

UI labels

Status color mappings

Static help text

Temporary controlled dropdown constants

Skeleton loaders

Empty states

Not Allowed

Fake equipment rows

Fake job request rows

Fake dashboard metrics

Fake vendors

Fake job cards

Fake calibration dates

Fake reports

If DB is Empty

Show zero count

Show “No records found”

Show empty state

Do not inject fake frontend data

21. Design Direction

You asked for a redesign prompt to make the entire project:

Classy

Glossy

Premium

Fully professional

Easy to use

Industry-grade

Government-grade

Minimal-click

Design Style

Light theme

Soft blue-grey background

Glossy white cards

Subtle glassmorphism

Rounded cards

Professional blue primary color

Slate typography

Calm status colors

Clean tables

Responsive forms

Premium command-center feel

Design Goal

Do not make it look like a hackathon project.

Do not make it look like generic admin dashboard.

Make it feel like a high-end engineering operations platform.

22. Suggested Project Folder Names

Production-Oriented

cmcmis-production

cmcmis-platform

cmcmis-system

cmcmis-enterprise

Design/Ease-Oriented

cmcmis-opsflow

cmcmis-smartflow

cmcmis-uxflow

cmcmis-ease

cmcmis-workbench

Recommended

cmcmis-opsflow

Reason:

It sounds production-grade, workflow-oriented, and focused on ease of operations.

23. Organizational / Infrastructure Context Draft

Organization

ISRO SAC-like high-precision engineering organization

The system is designed for technical/government-grade workflows involving equipment, calibration, repair, registration, procurement, and reports.

Internship Duration

45 days

So scope should be MVP-first, not full enterprise rollout.

Existing System

Existing database-backed legacy system

We are modernizing the application layer, not redesigning the database from scratch.

Deployment Assumption

Likely on-prem or organization-private infrastructure

Avoid assuming public cloud unless confirmed.

SSO

Future integration

Protocol not confirmed

Design must remain SSO-ready.

File Storage

Likely internal filesystem or network storage

For reports, documents, calibration certificates.

Notifications

In-app first

Email next

SMS/push later only if required.

Compliance

NABL / ISO 17025 readiness should be considered

Audit trail, report history, and traceability are important.

24. MVP Scope Recommendation

Version 1 Priority

1. Auth/RBAC
2. Dashboard
3. Equipment
4. Job Requests
5. New Job Request Form
6. Inquiry

Version 2 Priority

1. Job Cards
2. Scheduling
3. Reports
4. Procurement
5. Admin/Master Data
6. Real SSO

Why

V1 proves login, role permissions, real data retrieval, equipment visibility, request creation, and basic tracking.

25. End-to-End Real User Flows

Calibration Request Flow

User logs in

- Opens dashboard
- Clicks New Service Request
- Selects Calibration
- Searches existing equipment
- Equipment autofills
- Adds accessories if needed
- Submitted-by autofills
- Enters complaint/requirement
- Accepts T&C
- Submits request
- Request appears in Job Requests list
- Lab in-charge approves
- Job card created
- Engineer assigned
- Calibration performed
- Report uploaded
- Job completed

→ User tracks status

Repair Request Flow

User logs in

→ Creates Repair request

→ Selects equipment

→ Enters fault/complaint

→ Adds accessories if needed

→ Submits

→ Lab reviews

→ Engineer diagnoses

→ Spare/action added

→ Repair completed

→ Equipment returned

→ Request closed

Equipment Registration Flow

User clicks Add Equipment

→ Registration form opens

→ Enters equipment details

→ Enters procurement details

→ Adds accessories/manuals

→ Submitted-by autofills

→ Accepts registration T&C

→ Submits registration request

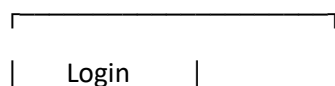
→ Lab verifies documents/equipment

→ Equipment ID assigned

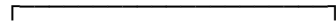
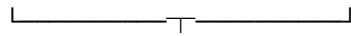
→ Equipment added to inventory

→ Calibration schedule created

26. Overall System Diagram

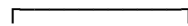
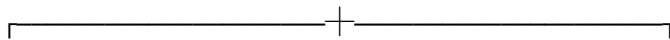
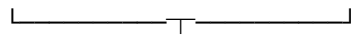


| Employee ID / SSO |

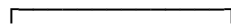


| Auth + RBAC |

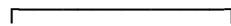
| Role + Permission |



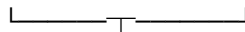
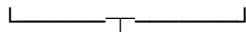
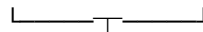
| Dashboard |



| Job Requests |



| Equipment |



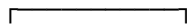
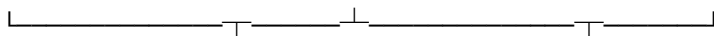
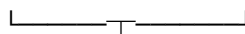
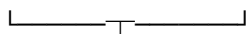
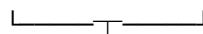
| Inquiry |



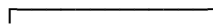
| New Request |



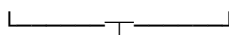
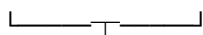
| Registration |



| Job Cards |



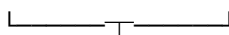
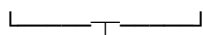
| Scheduling |

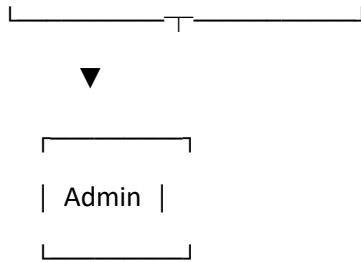


| Reports |



| Procurement |





27. Beginner-to-Advanced Concept Relationship

Concept	Simple Meaning	Technical Meaning
Auth	User login	Session + identity verification
RBAC	Role-based access	Permission middleware and scoped APIs
Dashboard	Overview page	Aggregated queries + cards/charts
Job Request	User request	Parent workflow record
Job Card	Work execution	Technical lifecycle record
Equipment	Asset/instrument	Inventory and calibration object
Inquiry	Search module	Read-only scoped lookup APIs
Schedule	Planned work	Calendar/table due-date workflow
Procurement	Purchase/warranty	PO, cost, vendor, spare tracking
Reports	Insights	Filtered exports and analytics
Admin	Control center	Master data + permissions

28. Final Project Vision

CMCMIS should become a complete smart operations platform where:

Users submit requests easily.

Lab engineers process work accurately.

Lab in-charges monitor and assign efficiently.

Admins control roles and master data.

Management gets reliable reports.

Equipment history remains traceable.

Calibration and repair workflows become transparent.

Final Qualities

Fast

Secure

Permission-aware

Minimal-click

Responsive

Database-driven

Production-ready

Government-grade

Maintainable

Auditable

Classy

Glossy

Professional

Easy to use

29. Response Style I Will Follow Now Onwards

From now on, explain everything in a structured step-by-step format using memory-based descriptions.

Include:

- table-wise explanations
- flowcharts
- diagrams
- architecture/design visuals
- relationships between concepts
- simple explanations + deep technical details
- real-world examples
- module connections
- summarized notes at the end

Make responses:

- visually organized
 - easy to revise
 - beginner-to-advanced friendly
 - practical for implementation
 - useful for prompts, documentation, debugging, and architecture
-

30. Short Summary Notes

CMCMIS = role-based calibration and maintenance operations platform.

Current stack = React + Vite + Tailwind + Express + MySQL.

Database = existing MySQL only, no new schema unless explicitly approved.

Current priority = Auth/RBAC → Dashboard → Job Requests → Equipment → Inquiry → New Request Form.

Registration = request flow, not direct equipment creation.

Add Equipment = opens registration request.

No mock operational data allowed.

Backend must enforce permissions.

Frontend must be responsive, classy, glossy, and easy to use.

Future = Job Cards, Scheduling, Procurement, Reports, Admin, SSO, Notifications, Audit, PDF certificates.